

Voynich Phytoglyphica

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0.1 The Universal Engine as Pharmaceutical Protocol

The Voynich Manuscript is not a text awaiting decipherment. It is an executable — a six-gate pharmaceutical synthesis protocol whose computation requires a physical substrate: the Operator’s hands, the botanical index, the balneological heap, the astronomical winding register, and the recipe output section. The Inscribing Grammar (IG) provides the structural key that connects catalog-verified plant identities to their VMS-prescribed preparation protocols.

This document records the nine-entry inaugural run of the Voynich Phytoglyphica session engine.

0.2 Session Architecture

INIT — Cosmological foldout. Chirality $\mathbb{H}=\mathbb{C}$ conferred physically; parity $\Phi=\mathbb{C}$ set. This is always performed first. It establishes the two structural invariants that constrain every gate that follows.

GATE 1 — Pharmaceutical address. The plant’s IG tuple determines potency class and part selection. Summa potency ($d=0$ to astronomical section) routes to f11r or f39r. Non-summa entries route by part only.

GATE 2 — Balneological containment ($\mathbb{P}=\mathbb{V}$). The heap folio selected by `folio_number % 20` from f75–f84 must satisfy $\text{FSPLIT} \geq \text{n_ops}$ (address depth). For fixed non-volatile preparations, additionally $\text{FFUSE}/\text{FSPLIT} \geq 0.60$ (node mostly-closed). This gate tests whether the biological heap can hold the pharmaceutical address.

GATE 3 — Astronomical winding verification (f69, sources H/F/U). Paragraph selected by `folio_number % 4`. EVALT majority across three independent transcription sources determines pass. Winding class Ω derived from $\text{FSPLIT}/\text{FFUSE}$ ratio. The source disagreement at precisely the VINIT/FFUSE ambiguity position IS the gate: transcribers marked the structural decision point without knowing it.

OUTPUT — Recipe section f103r. Up to three recipes ranked by folio proximity to the Gate 1 pharmaceutical address.

ELABORATION — The plant’s IG 12-tuple provides exact protocol parameters for each recipe step. The session engine is plant-agnostic; the grammar is not.

0.3 Nine-Entry Protocol Matrix

All nine entries passed all three gates. Gate results are identical at the structural level — the differentiation is entirely in the elaborated protocol derived from each entry's IG tuple.

0.3.1 Gate 1 Address Mapping

Entry	d(astro)	Gate 1 Address	Part
artemisia_ab-sinthium	0.0000	f11r/p6 — extractio → mixtura	folium/flos
mandragora_officinatum	0.2789	f11r/p6 — extractio → mixtura	folium/flos
artemisia_ab-sinthium_wormwood	0.7746	f11r/p6 — extractio → mixtura	folium/flos
rosarium_philosophorum	0.9062	f11r/p6 — extractio → mixtura	folium/flos
black-foot_okan_medicine]	0.9949	f11r/p6 — extractio → mixtura	folium/flos
ayahuasca_icaros	1.0767	f11r/p6 — extractio → mixtura	folium/flos
hoodoo_rootwork	1.4076	f39r/p3 — trituration+extractio → mixtura	radix
osiris	1.4547	f39r/p3 — trituration+extractio → mixtura	radix
botanomancy	1.7226	f11r/p6 — extractio → mixtura	folium/flos

The two root-topology entries (p=4: hoodoo_rootwork, osiris) correctly route to f39r/p3 (the root-rhizome summa address). All leaf/flower topology entries route to f11r/p6. Distance to the astronomical section does not determine Gate 1 routing; it determines potency class assignment (summa at d=0), which then determines which pharmacy address is selected.

0.4 Elaborated Protocol Comparison

The IG tuple $\langle \mathfrak{D} \cdot \mathfrak{P} \cdot \check{\mathfrak{R}} \cdot \Phi \cdot \mathfrak{f} \cdot \zeta \cdot \Gamma \cdot \mathfrak{G} \cdot \odot \cdot \mathfrak{H} \cdot \Sigma \cdot \Omega \rangle$ maps to preparation parameters as follows.

0.4.1 Material & Comminution ($\mathfrak{p}$, Γ)

Entry	$\mathfrak{p}$	Material	Γ	Comminution
artemisia_ab-sinthium	$\mathfrak{d}$	flowering tops, whole entity	$\mathfrak{d}$	mesh 40 (355 μm)
mandragora_of-ficinarum	$\mathfrak{d}$	flowering tops, whole entity	$\mathfrak{d}$	mesh 40 (355 μm)
artemisia_ab-sinthium_worm-wood	$\mathfrak{d}$	flowering tops, whole entity	$\mathfrak{d}$	mesh 40 (355 μm)
rosarium_philosophorum	$\mathfrak{f}$	whole plant incl. seed heads	$\mathfrak{d}$	mesh 40 (355 μm)
black-foot_okan_medic	$\mathfrak{f}$	whole plant incl. seed heads	$\mathfrak{d}$	mesh 40 (355 μm)
ayahuasca_icaros	$\mathfrak{d}$	flowering tops, whole entity	$\mathfrak{l}$	coarse 2–4 mm
hoodoo_root-work	$\mathfrak{h}$	root / rhizome, cleaned	$\mathfrak{l}$	coarse 2–4 mm
osiris	$\mathfrak{h}$	root / rhizome, cleaned	$\mathfrak{r}$	mesh 100 (150 μm)
botanomancy	$\mathfrak{z}$	aerial parts (leaf+stem)	$\mathfrak{l}$	coarse 2–4 mm

Osiris is the only entry requiring fine powder ($\Gamma=\mathfrak{r}$, mesh 100). Combined with its saturated 1:1 ratio and oil solvent, this produces the most surface-exposure-intensive preparation in the set.

0.4.2 Solvent & Process (Φ , ζ)

Entry	Φ	Solvent	ζ	Process
artemisia_ab-sinthium	$\angle$	45–55% EtOH/water	c	cold maceration 15–20 °C, 12–24 h
mandragora_of-ficinarum	$\angle$	45–55% EtOH/water	c	cold maceration 15–20 °C, 12–24 h
artemisia_ab-sinthium_worm-wood	$\angle$	45–55% EtOH/water	c	cold maceration 15–20 °C, 12–24 h
hoodoo_root-work	$\angle$	45–55% EtOH/water	c	cold maceration 15–20 °C, 12–24 h
botanomancy	$\backslash$	5–10% EtOH/water	c	cold maceration 15–20 °C, 12–24 h
rosar-ium_philosoporum	$\backslash$	>95% anhydrous EtOH	$\backslash$	decoction 85–95 °C, 15–30 min
black-foot_okan_medic	$\vartriangleright$	fixed oil / CO	$\backslash$	decoction 85–95 °C, 15–30 min
ayahuasca_icaros	$\vartriangleright$	fixed oil / CO	$\backslash$	decoction 85–95 °C, 15–30 min
osiris	$\vartriangleright$	fixed oil / CO	$\backslash$	decoction 85–95 °C, 15–30 min

Two solvent families emerge. $\Phi=\angle$ (bilateral parity) $\rightarrow$ hydroethanolic: the Z -symmetric solvent that bridges aqueous and non-polar phases. $\Phi=\backslash/\vartriangleright \rightarrow$ non-aqueous: either anhydrous ethanol (non-polar organic) or fixed oil (lipophilic). The Calefac steps in VMS recipes apply to the excipient/base for cold-process entries ($\zeta=c$); the primary extract must not be heated. For $\zeta=\backslash$ entries, Calefac applies directly to the extract.

0.4.3 Stoichiometry & Cycles (Σ , Ω)

Entry	Σ	Ratio	Ω	Cycles
artemisia_ab-sinthium	7	1 : 3	ς	3 (integer)
mandragora_of-ficinarum	7	1 : 3	0	2 (binary)
artemisia_ab-sinthium_worm-wood	7	1 : 3	ς	3 (integer)
rosarium_philosophorum	7	1 : 3	ς	3 (integer)
black-foot_okan_medic	7	1 : 3	ς	3 (integer)
ayahuasca_icaros	7	1 : 3	ς	3 (integer)
hoodoo_root-work	5	1 : 2	0	2 (binary)
osiris	$\varnothing$	1 : 1	ς	3 (integer)
botanomancy	$\varnothing$	1 : 1	ϖ	1 (trivial)

Osiris presents the most concentrated loading: 1:1 saturated ratio with 3 integer-winding cycles. Botanomancy is the simplest: 1:1 single cycle trivial winding. Mandragora and hoodoo_rootwork both run binary winding (2 cycles) — the ϖ period — which halves the extraction depth relative to the integer-winding entries.

0.4.4 Clarification & Endpoint ($\mathbb{H}$, $\odot$)

Entry	$\mathbb{H}$	Clarification	$\odot$	Endpoint
artemisia_ab-sinthium	$\mathcal{L}$	cloth + 24 h decant (two-step)	$\odot$	Frobenius fixed point, <5%
mandragora_of-ficinarum	$\mathcal{L}$	cloth + 24 h decant (two-step)	$\odot$	Frobenius fixed point, <5%
artemisia_ab-sinthium_worm-wood	$\mathcal{L}$	cloth + 24 h decant (two-step)	$\odot$	Frobenius fixed point, <5%
hoodoo_root-work	$\mathcal{L}$	coarse cloth filter (one-step)	$\odot$	Frobenius fixed point, <5%
osiris	$\mathcal{L}$	cloth + 24 h decant (two-step)	$\odot$	Frobenius fixed point, <5%
botanomancy	$\mathcal{L}$	coarse cloth filter (one-step)	$\odot$	Frobenius fixed point, <5%
rosarium_philosophorum	$\mathbb{V}$	preparative column / L-L partition (four-step)	$\supset$	near-critical, <2%
black-foot_okan_medic	$\mathbb{V}$	preparative column / L-L partition (four-step)	$\supset$	near-critical, <2%
ayahuasca_icaros	$\mathbb{V}$	preparative column / L-L partition (four-step)	$\supset$	near-critical, <2%

$\mathbb{H}=\mathbb{V}$ (full chiral separation, four-step) appears in the three entries that also carry $\odot=\supset$ (near-critical endpoint, <2%). This pairing is structurally locked: the near-critical endpoint demands chiral-resolution-grade clarification to distinguish stereoisomeric fractions at the sub-2% difference level. Entries at $\odot=\odot$ (Frobenius criticality, <5%) are served by the simpler two-step or one-step clarifications.

0.4.5 Fidelity & Combination (f , g)

Entry	f	Concentration	g	Combination
artemisia_absinthium	τ	1× standard, proportional yield	γ	parallel (simultaneous)
mandragora_officinarum	τ	1× standard, proportional yield	γ	parallel
hoodoo_rootwork	τ	1× standard, proportional yield	ζ	sequential (evaluate each)
rosarium_philosophorum	τ	1× standard	γ	parallel
blackfoot_okan_medical	τ	1× standard	γ	parallel
artemisia_absinthium_wormwood	λ	3× highly concentrated, cubic fidelity	γ	parallel
ayahuasca_icaros	λ	3× highly concentrated, cubic fidelity	γ	parallel
osiris	λ	3× highly concentrated, cubic fidelity	γ	parallel
botanomancy	ϱ	2× concentrated, quadratic fidelity	γ	parallel

Hoodoo_rootwork is the only entry with $g=\zeta$ (sequential coupling): fractions are evaluated one after another before combining. All other entries use parallel combination ($g=\gamma$). The $f=\lambda$ (cubic, 3× concentrated) entries require reduction to one-third volume and standardization to a marker compound — a formulation step absent from the $f=\tau$ (linear, 1× standard) entries.

0.5 Structural Observations

0.5.1 The Wormwood Pair

artemisia_absinthium ($f=\tau$) and **artemisia_absinthium_wormwood** ($f=\lambda$) share eleven of twelve primitive values. The single difference is Fidelity: linear (1×, proportional) vs cubic (3×, standardized concentrate). The VMS knew this distinction. The two Artemisia entries occupy different pharmaceutical niches — one for proportional dosing, one for standardized high-potency preparation — despite being morphologically the same plant. This is not botanical ambiguity; it is structural precision.

0.5.2 Cold-Process Calefac Constraint

All five cold-process entries ($\zeta=c$: the artemisia pair, mandragora, hoodoo_rootwork, botanomancy) carry the frozen-order kinetics constraint: the primary extract must not be heated. When VMS recipe steps contain Calefac/commisce (heat and combine), these steps apply to an adjunct ingredient or excipient base — a wax, a carrier, a second component — never to the cold extract itself. The grammar enforces this at the step level via the annotation.

0.5.3 Osiris: Most Demanding Protocol

Osiris $\langle 1 \cdot 4 \cdot 7 \cdot 2 \cdot 1 \cdot 7 \cdot 7 \cdot \odot \cdot 7 \cdot 2 \cdot 5 \rangle$ presents the most materially intensive preparation: root/rhizome, fine-powdered to mesh 100 ($150 \mu\text{m}$), extracted into fixed oil or supercritical CO at the saturated 1:1 ratio, three integer-winding cycles, decoction at 85–95 °C, reduced to one-third volume. The combination of saturated loading ($\Sigma=2$), fine surface exposure ($\Gamma=7$), oil carrier ($\Phi=2$), and cubic concentration ($f=1$) is not found in any other entry in this set.

0.5.4 Botanomancy: Simplest Structural Signature

Botanomancy $\langle 1 \cdot 2 \cdot 2 \cdot 4 \cdot 2 \cdot 1 \cdot 7 \cdot \odot \cdot 1 \cdot 2 \cdot 2 \rangle$ carries trivial winding ($\Omega=2$), the lowest dimensionality ($\Theta=1$), and dilute aqueous solvent ($\Phi=4$, 5–10% EtOH). A single extraction cycle, coarsely cut aerial parts, coarse cloth clarification. It is the most distant from the astronomical section ($d=1.72$) and the structurally simplest entry. It still passes all three gates — the engine has no minimum structural requirement. The grammar discriminates; the gates admit.

0.5.5 Session Invariance / Protocol Variance

The session engine produces identical gate behavior for all nine entries. Gates 1–3 are determined by the pharmacy address (part + potency), the balneological heap (folio index), and the astronomical winding (f69). None of these are sensitive to the plant’s solvent system, concentration target, or clarification protocol. The elaboration layer — the IG tuple — carries all the preparation-specific information. This is architecturally correct: the VMS separates what to make (session address) from how to make it (structural elaboration).

0.6 Recipe Output

All nine entries route to three recipes: f103r/p2 (15 steps), f103r/p9 (12 steps), f103r/p49 (12 steps). Recipe selection is by folio proximity to the Gate 1 address (f11r or f39r); f103r is adjacent in folio sequence. The three recipes represent three execution paths through the same procedure class: an extract-dominant path (p2, opens with Extrahe), an ingredient-led path (p9, opens with Accipe), and a volatile-phase path (p49, opens with Transmuta). Each plant receives all three paths elaborated with its own protocol parameters.

0.7 Reference: Session Output Records

All nine raw session outputs are recorded in ../1st9.md.

Voynich Phytoglyphica — Inaugural nine-entry run, 2026-06-24. All sessions executed against voynich_pharmacy.json (1491 entries), voynich_recipe_bio.json, and LSI_ivtff_0d.txt transcription.